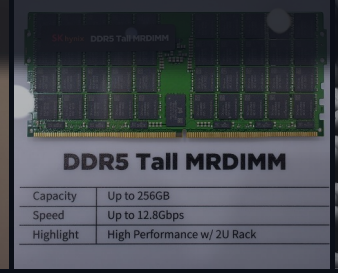
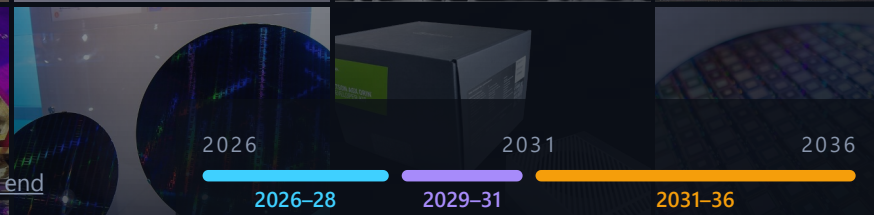
SEMICONDUCTORS 2026 → 2036

# After the Shrink

Raveh Neeman · October 21, 2026

The roadmap of the semiconductor industry

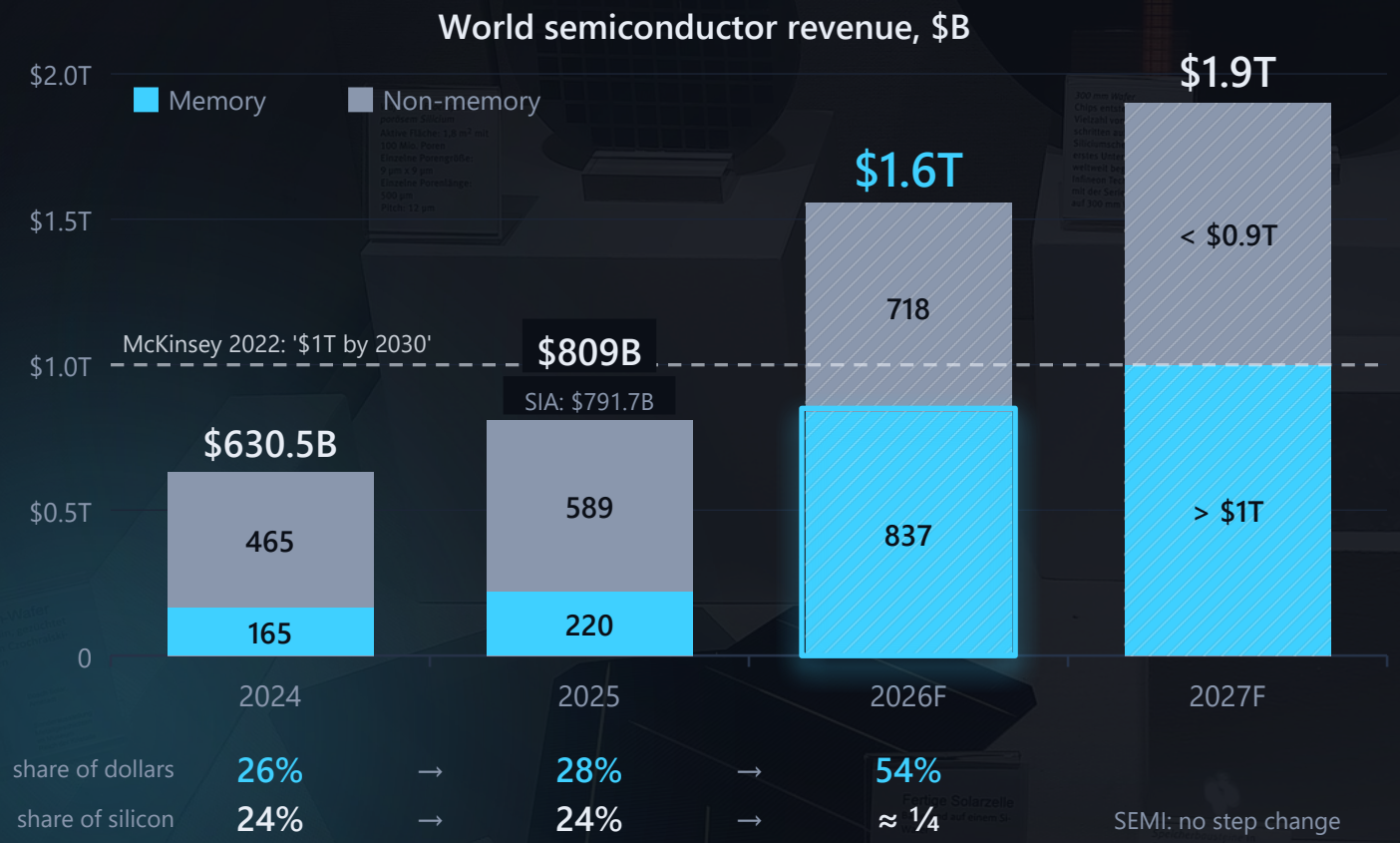



IRDS/ISRDS 2026 workshops, Granada — sinano.eu/irds-isrds-presentations · Images: Wikimedia Commons (PD, CC0, CC BY) — credits at the end

THE MARKET · 2026

# 2026: the market doubles — on memory prices




**$1T**
four years early

**54%**
of the money (2026F)

**24%**
of the silicon area

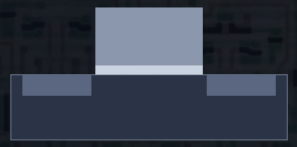
Bits bought: **+20% a year**. Price per bit 2026: **×3**.

**The difference is price — and memory booms revert: 2019, 2023**

SIA Feb 2025 · SIA/WSTS Feb & Sep 2026 · Gartner 24 Aug 2026 (forecast) · SEMI World Fab Forecast Jan 2024 · Jan 2025 · SEMI 300mm Fab Outlook Jun 2026 · Micron calls 2022–25 · McKinsey 2022

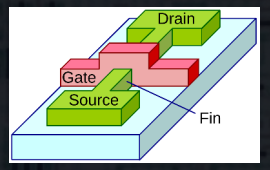
THREE ERAS OF SCALING

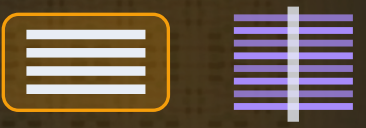
# Scaling changed its goal: from speed to energy



~2005: clock stops rising

$$P \approx N \cdot C \cdot V^2 \cdot f \cdot A$$



3D power scaling → 2040

2003 2007 2011

**Geometrical scaling**
~1975–2003 · shrink every dimension

**Equivalent scaling** ~2003–2024
strain 2003 · high-k/metal gate 2007 · FinFET 2011

memory **2015** (3D NAND)

logic **2025** (GAA nanosheets)

1970 1980 1990 2000 2010 2020 2030 **2040**

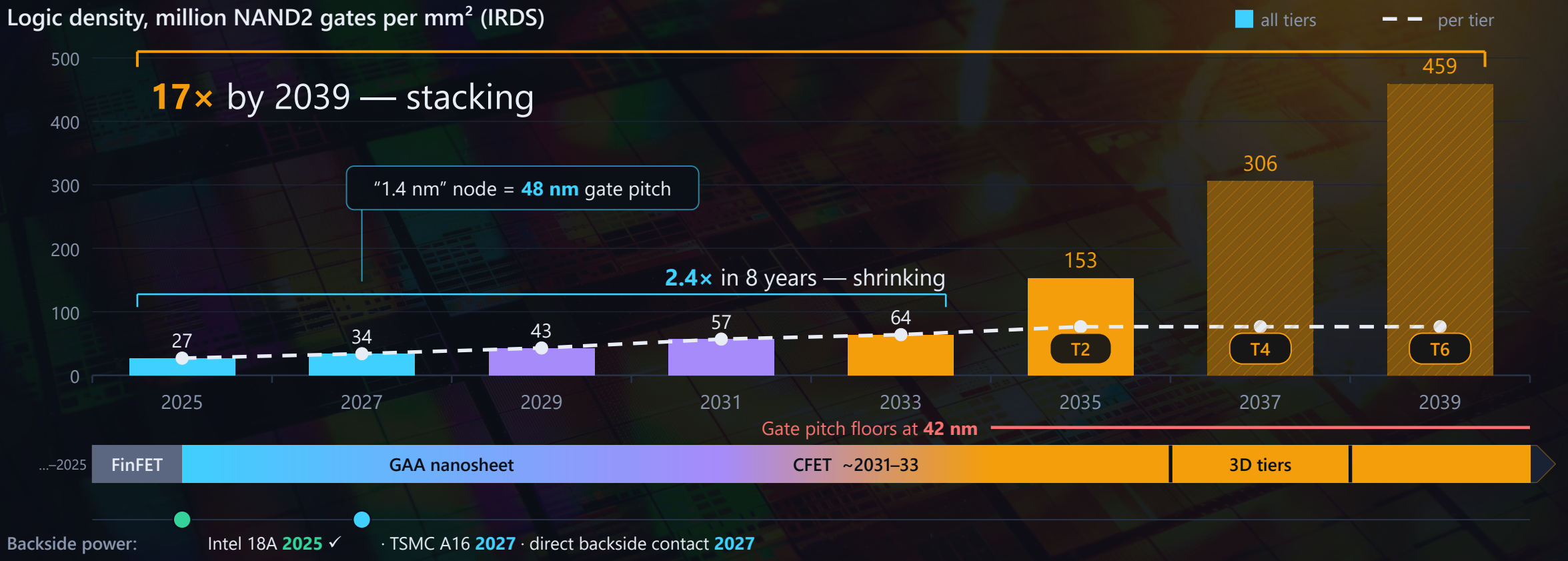
> *"Minimizing switch energy was the new rule."*
> — IRDS 2024 Prologue

2026 2031 2036 2040

Gargini, IRDS Granada 2026 (Intro p.14, p.16) · IRDS 2024 Prologue & Exec. Summary p.7 · Gargini, ConFab 2015

LOGIC ROADMAP 2025–2039

# Density now grows up, not across




IRDS More Moore, Granada 2026 (Badaroglu p.6, 7, 16) & 2024 Table MM01 · IRDS Litho 2025 · imec 2026 roadmap · TSMC, Intel

2026 2031 2036

COST OF THE LEADING EDGE

# The shrink got expensive, and gives back less

**$380–400M**
High-NA EUV scanner

Intel: in production

TSMC: not through 2029

**≈$30k**
2 nm wafer (reported)

**≈$725M**
2 nm design (estimate)

Density gain per node: **1.9×** (2017) → **1.23×** (2028)




Cost per transistor: flat since **28 nm**



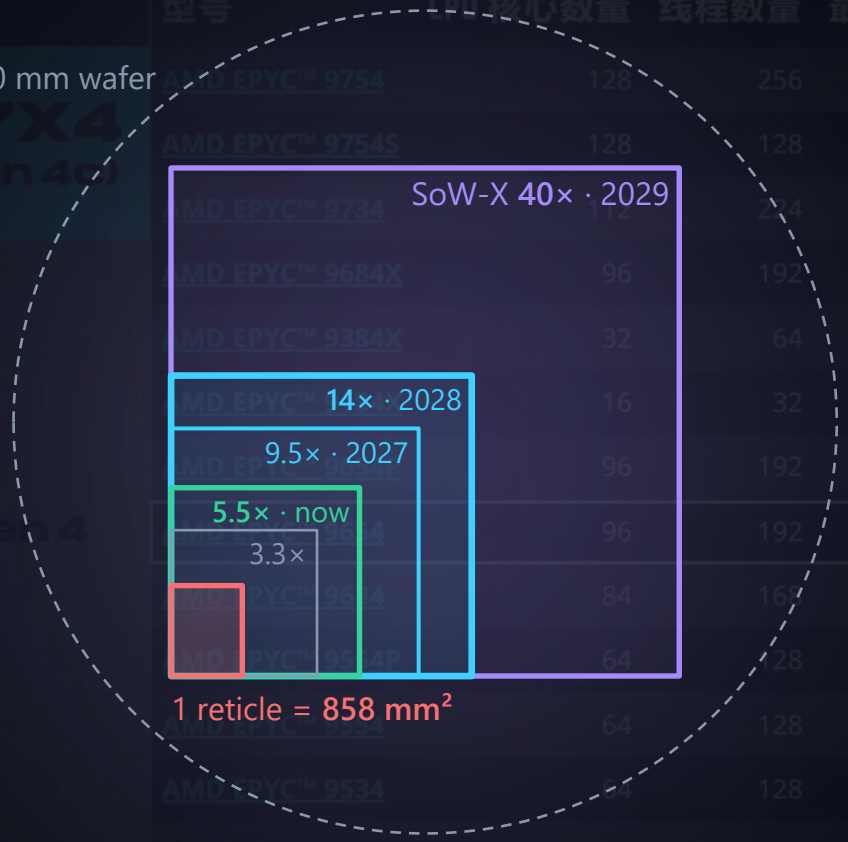

## Node migration no longer pays for itself.




ASML Q2 2026 call; Tom's HW (Jul 2026) · TSMC Symposium Apr 2026 · EE Times · IBS via CSIS (Sep 2026) · Google/IEDM via Tom's HW · IRDS Litho 2026

PACKAGING · 2026 → 2031

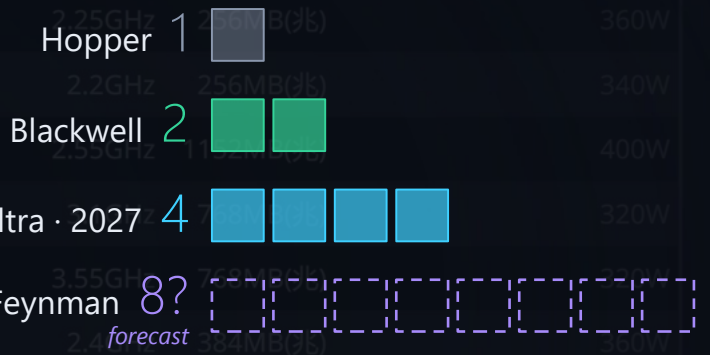
# The package is the new chip






GPU + HBM on a silicon interposer

**GPU dies per package**




Die-to-die energy:
SerDes **5–10 pJ/bit** → 3D bond **0.01–0.05 pJ/bit**

Hybrid-bond pitch 9 → 6 → 4.5 µm

**Open chiplet market: not yet**

## Gargini's law: bits per package double every 2 years



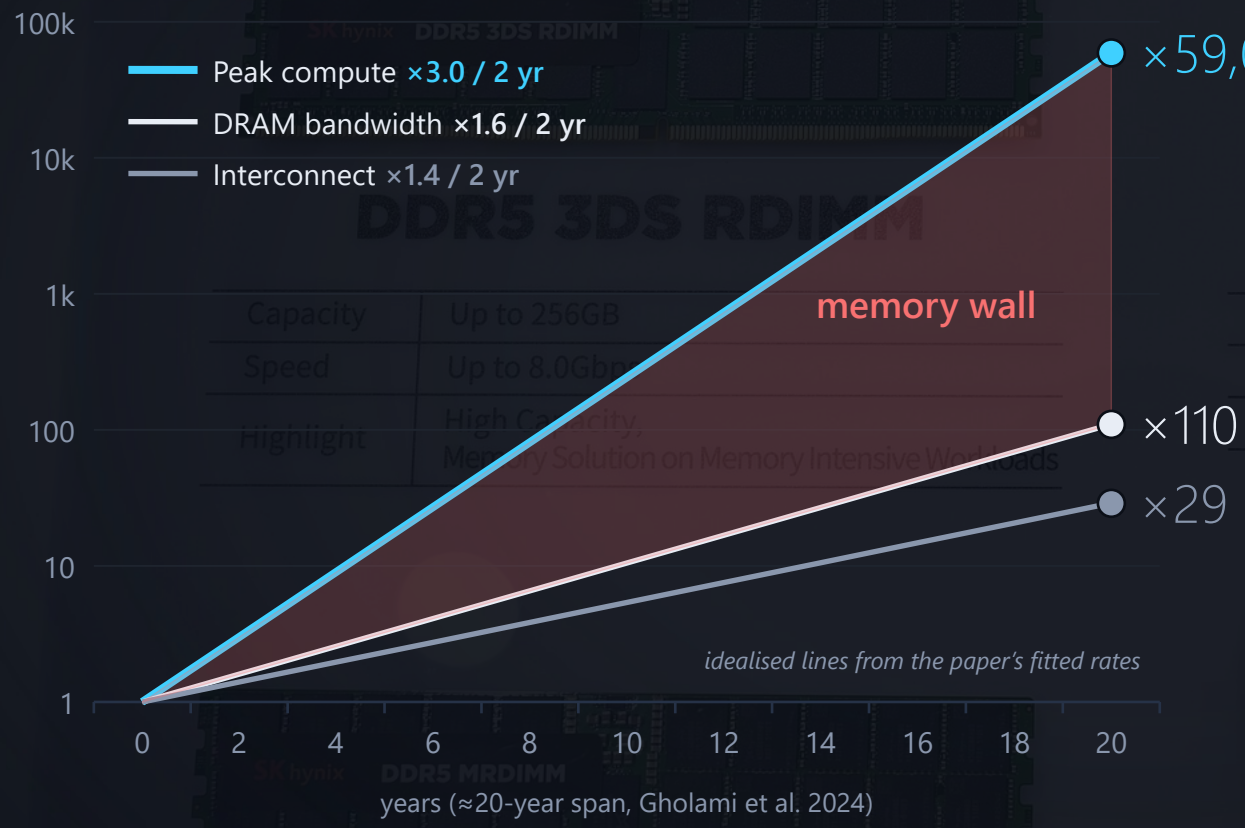

TSMC Technology Symposium 2026 · Gargini, IRDS/ISRDS Granada 2026 · UCIe 2.0 · IRDS 2024 SA · Photo: Fritzchens Fritz, CC0

MEMORY · 2026 → 2036

DRAM access 2–3 pJ/bit · compute ~10 fJ/bit

# Throughput: the memory and interconnect wall

Growth over ~20 years (index, log scale)




**200–300×**
energy: fetch a DRAM bit vs compute on it

**2048-bit** HBM4 bus
**2.8–3.3 TB/s** per stack, shipping 2026

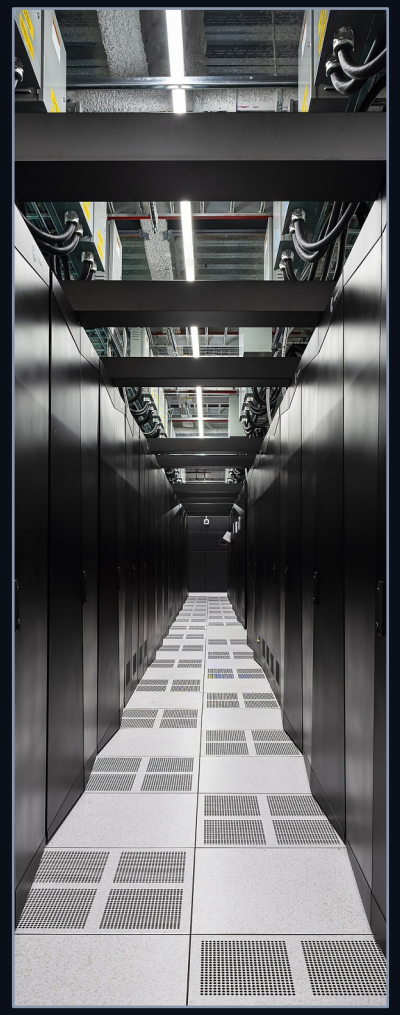
HBM base die → foundry logic (2027)

**+93–98%**
DRAM contract price, one quarter (1Q26)

## Design for bytes moved, not FLOPs




Gholami et al., IEEE Micro 2024 · IRDS More Moore 2026 · JEDEC · Samsung · Micron · TrendForce

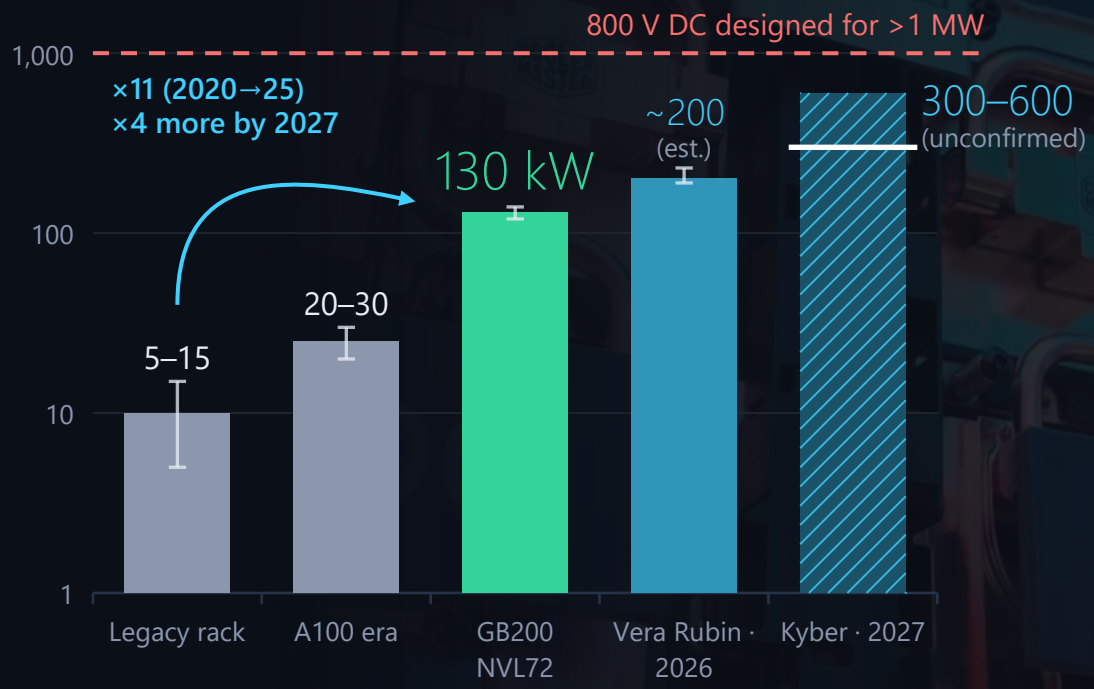


Frontier, ORNL · water-cooled · 21 MW

SYSTEMS · 2026 → 2028

# The rack is the new computer

**Power per rack (kW, log scale)**




45 °C
water, no chillers

At 54 V, 1 MW = 18.5 kA
and up to **200 kg** copper

30 W → 9 W
per optical port (co-packaged optics, vendor)

485 → ~950 TWh
data-centre electricity, 2025→2030

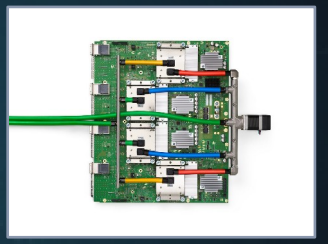
*Racks in data centers have become like transistors in ICs.*

2026 2031 2036

Gargini, IRDS/ISRDS Granada 2026 · IEA Energy & AI · NVIDIA (800 V DC, CPO) · DCD · Photos: Geekerwan, CC BY 3.0 · Oak Ridge National Laboratory, CC BY 2.0

CUSTOM SILICON · 2026 → 2031

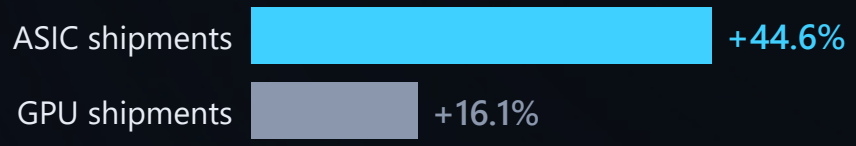
# Every AI giant now designs its own silicon



TPU v4: ASIC + 4 HBM




AI servers shipped, 2026




| | | |
|---|---|---|
| **Google** | Ironwood TPU v7 | 4.6 PF FP8 |
| **AWS** | Trainium3 | 3 nm |
| **Microsoft** | Maia 200 | >10 PF FP4 |
| **Anthropic** | own chip team (Aug 2026)<br>buys TPU · Trainium · NVIDIA · AMD | 10+ GW<br>announced |
| **OpenAI + Broadcom** | Jalapeño · ~840 mm² | 9 months<br>to tape-out |

FPGA: Lattice +42% YoY, AI servers

RISC-V: ~32% of SoCs by 2031 (forecast)

Emulation & prototyping: **record demand**

## Demand shifts to architects, verification and prototyping engineers




TrendForce via Tom's Hardware · Google, AWS, Microsoft, Broadcom · Anthropic via DCD · Lattice · SHD Group · IRDS 2024 SA · Photo: Jouppi et al., CC BY 4.0

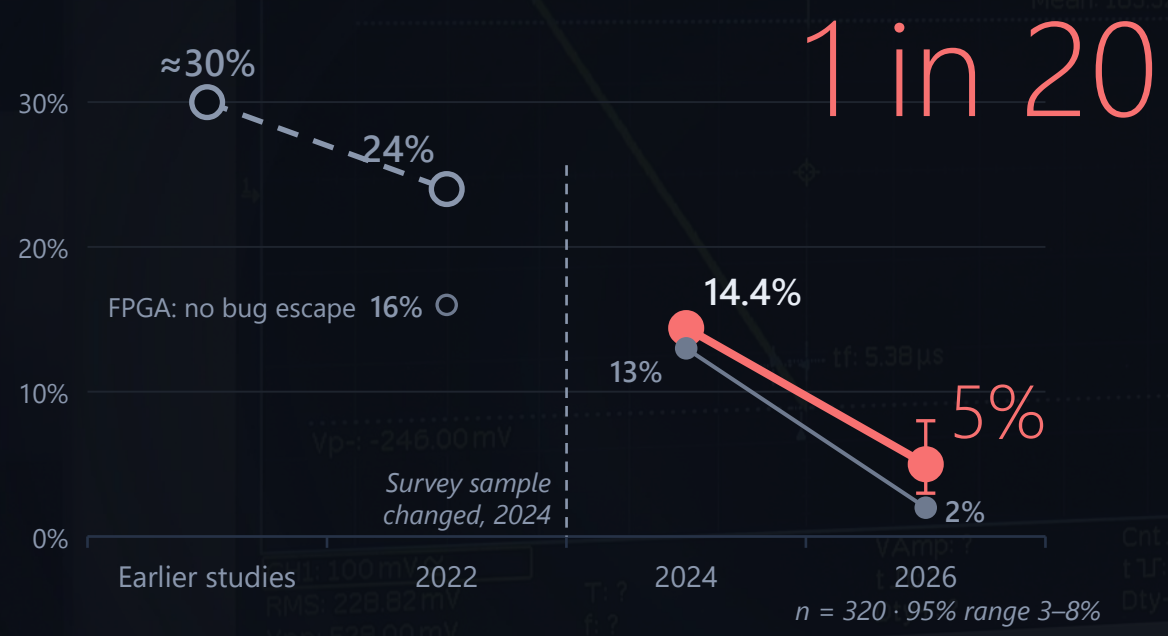
VERIFICATION · 2026 → 2028

9-month tape-outs

# Reported first-silicon success has fallen to 1 in 20

IC/ASIC survey respondents · any spin before production counts, planned or not

## First-silicon success (IC/ASIC projects)

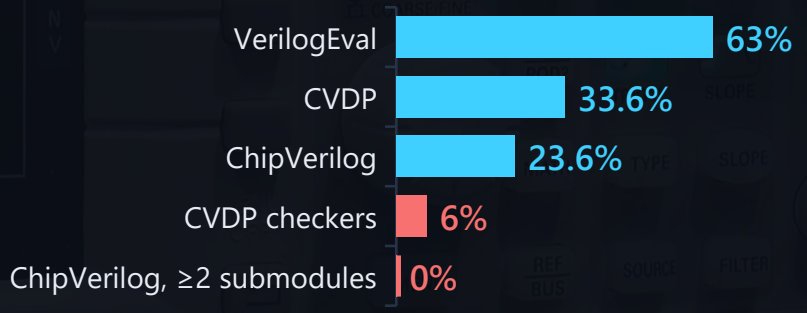



## Marketing vs measured

Vendor "up to" speed-ups




Measured pass@1




75% of projects late (2024)

49% of project time in verification (2026)

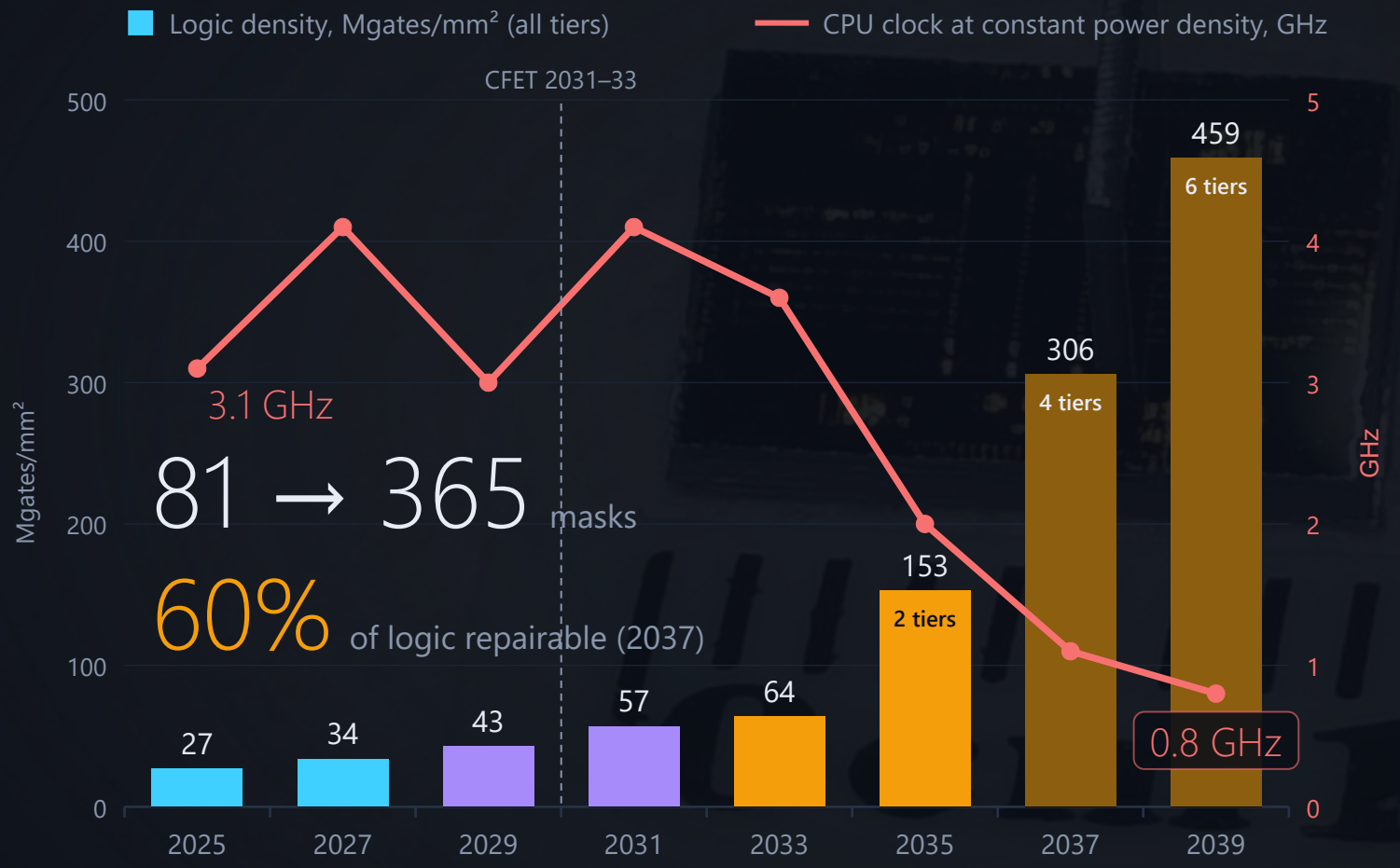
**Agents write stimulus. Humans own intent, spec, (reference model and checkers ?).**

2026 2031 2036

Siemens EDA / Wilson Research Group 2026 & 2024 · survey method · Cadence · Synopsys · CVDP (NVIDIA) · ChipVerilog

THE 3D ERA

# After 2031, chips stack up and clock down




**ARRIVES**

- eMRAM · eRRAM
- Analog in-memory (2.5–20 TOPS/W)
- Optical I/O chiplets

**STAYS IN THE LAB · to 2036**

- Steep-slope switches
- Spin logic
- 2D-channel logic

**No beyond-CMOS switch beats CMOS. A rival needs ~10×.**




IRDS 2024 More Moore Table MM01 · IRDS Beyond-CMOS 2024 · imec 2026 roadmap

ISRAEL · THE PLANS

# Israel after the shrink: design nation

### NVIDIA · Kiryat Tivon

>10,000

people in a new R&D campus, 160,000 m²; occupancy 2031. About 5,000 in Israel today

land bought Dec 2025

### Intel · Kiryat Gat

$25 B

Fab 38 expansion plan, $3 B of it state aid — frozen since July 2025; Fab 28 keeps running

no restart announced

### Tower · Migdal HaEmek

$1 B

silicon-photonics revenue run rate by Q4 2026; record $460 M quarter; 2028 target $3.6 B

new 300 mm fab: Japan

### Innovation Authority + MAFAT

NIS 150 M

national photonic-chip R&D facility, operating within 18 months

call opened Jun 2026

### National AI Directorate

100,000

GPUs in a cluster within five years, and a leading-edge fab — reported talks with TSMC, Samsung, Intel

reported, not signed

**45,000** people in **250+** chip companies — and a senior hardware hire takes **5–7 months**. The constraint is people, not money.



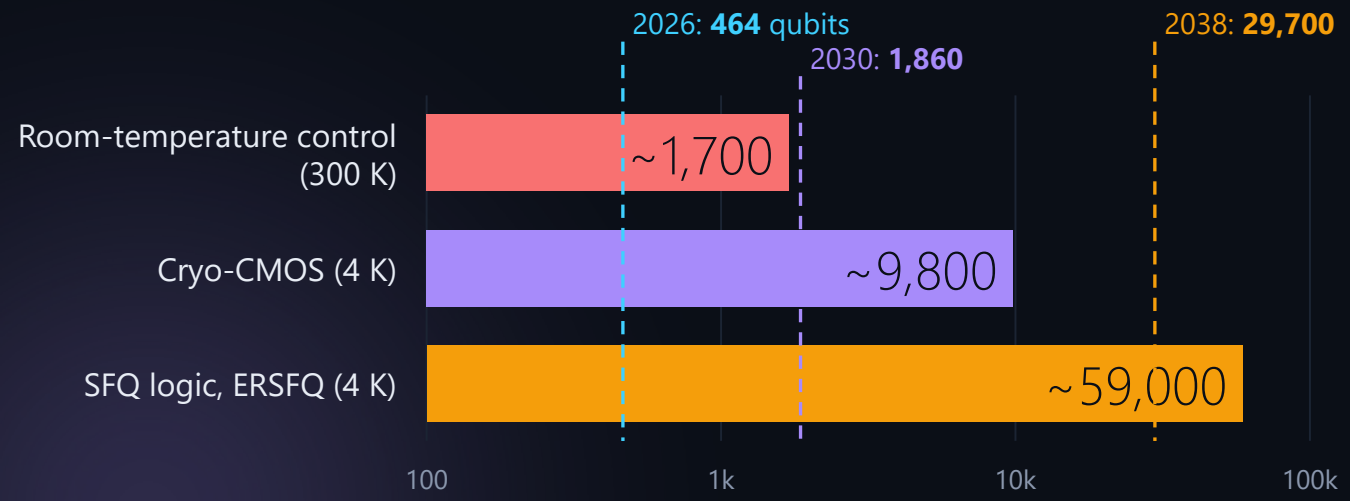

Globes 10 Jul 2025 · CTech 18 Dec 2025 · Tower Q2 2026 · IIA/MAFAT Jun 2026 · Startup Nation Central 2025 · Globes 10 Jun 2026

COLD COMPUTING

# Quantum's bottleneck is classical electronics

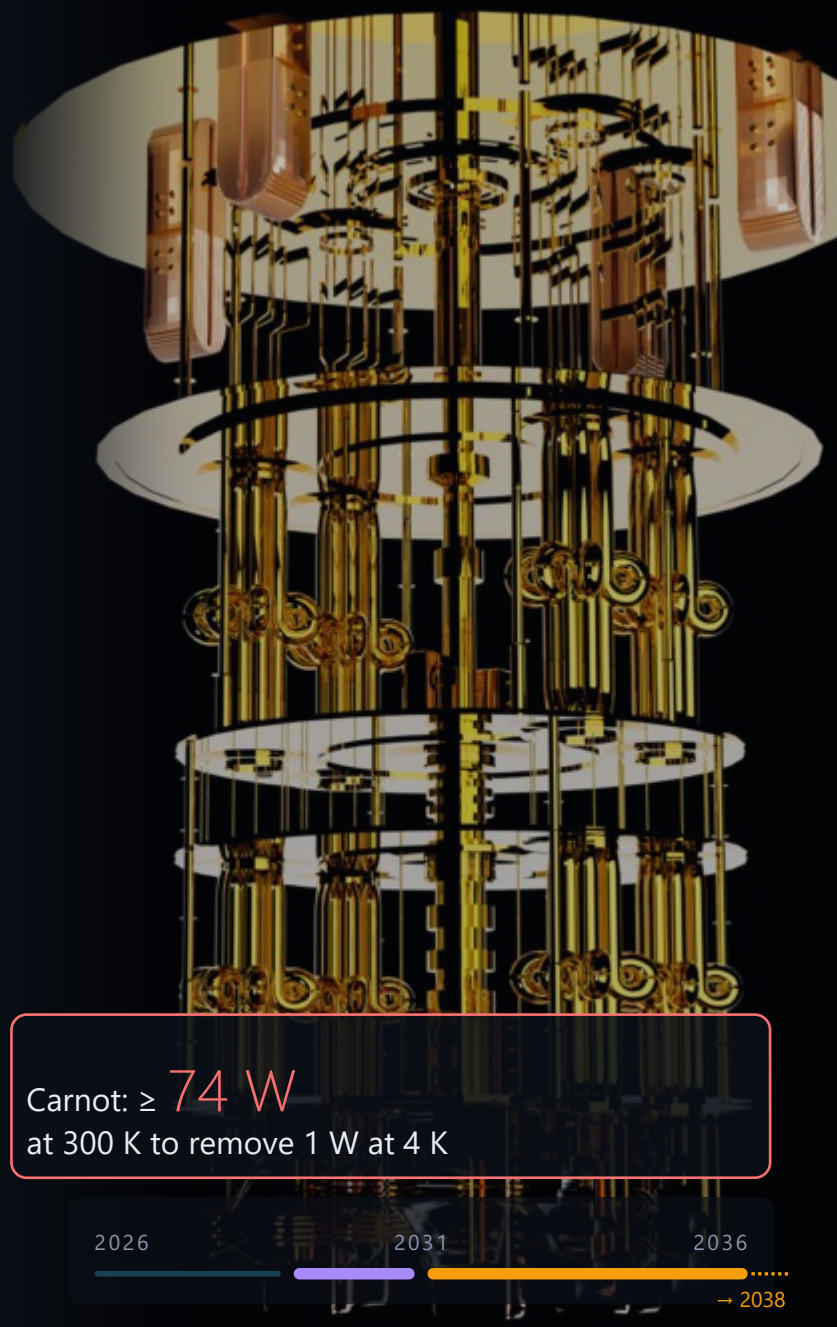
300 K → 4 K CMOS ~2030 → Superconducting ~2038

**Qubits one 4 K stage can control (IRDS model: 1.5 W, 620 cm²)**




## < 480 ns

mean time to decode 12 cycles on FPGAs (IBM, simulated data)



Carnot: ≥ 74 W
at 300 K to remove 1 W at 4 K

2026 — 2031 — 2036 → 2038

IRDS 2026 CEQIP (Cryogenic Electronics & Quantum Information Processing) · IBM Quantum, Nov 2025

PART 2

# Quantum computers update

Images: credits at the end



THE REASON

# Why quantum computers? Because only they can…

## 1 Simulate quantum matter and fields

drug metabolism (cytochrome P450), $CO_2$ catalysts, cuprate superconductors, battery cathodes, nuclei

Feynman 1982
Lloyd 1996

## 2 Quantum networks and quantum sensors — one ecosystem

entanglement distribution and quantum key distribution; sensors and clocks whose quantum data a processor learns from exponentially faster (Science 2022)

Kimble 2008
Degen et al. 2017

## 3 Quantum neural networks — a speculation

networks whose weights and activations are quantum states; no proven edge on ordinary data yet, barren plateaus in the way — my own bet, not a theorem

speculation
R. Neeman 2026

## 4 Break today's public-key cryptography

RSA and elliptic curves; the US orders post-quantum key exchange for high-value systems by 2030

Shor 1994
US deadline 2030

Missing on purpose: search (only quadratic), optimisation, AI on ordinary data (dequantised); sampling and structured algebra — no buyer yet.

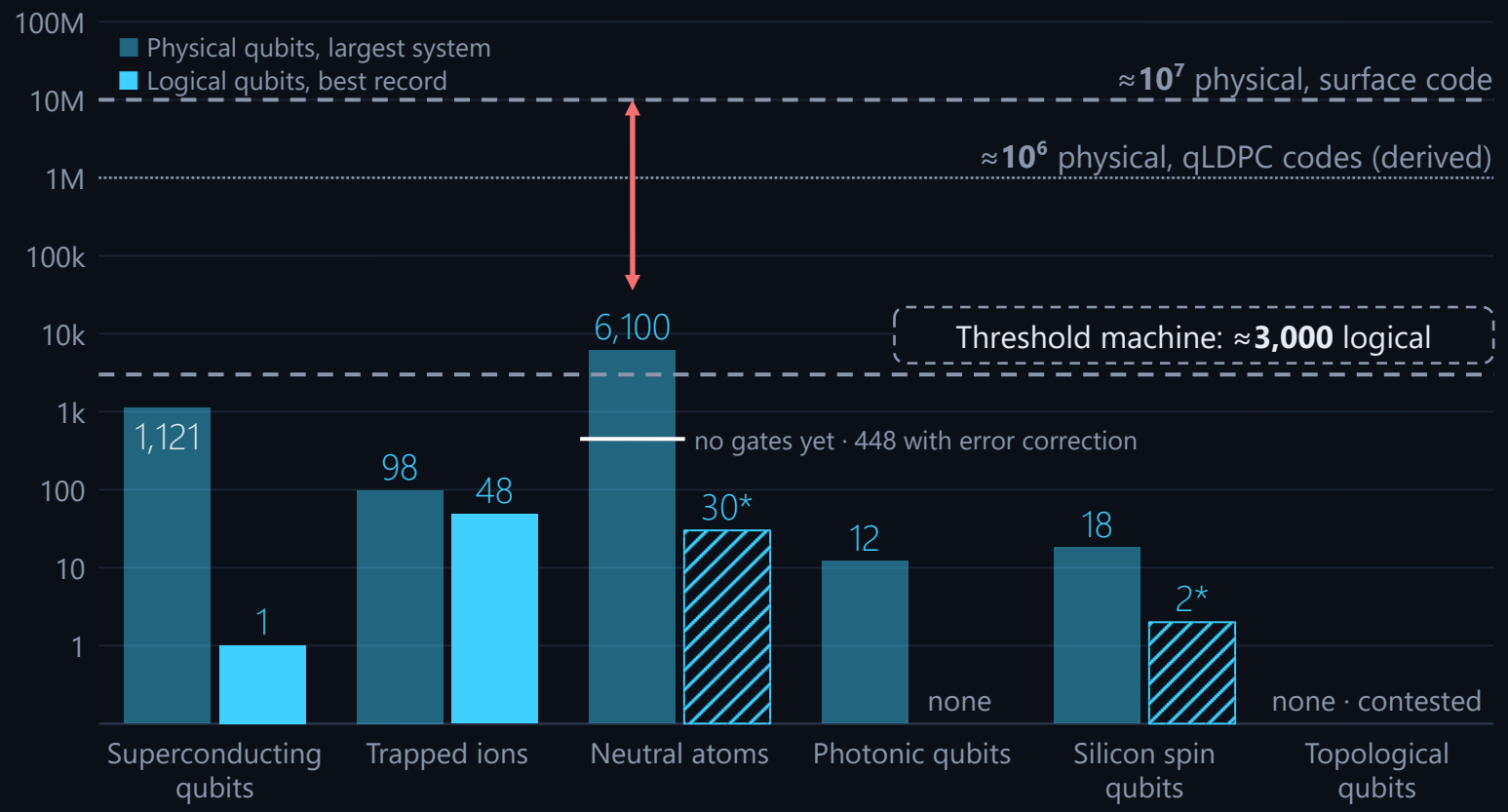



Feynman 1982 · Lloyd 1996 · Kimble 2008 · Degen, Reinhard, Cappellaro 2017 · Huang et al., Science 2022 · Larocca et al. 2025 (barren plateaus) · Shor 1994 · US EO 14412 (2026)

THE GAP

# Today: dozens of logical qubits. Needed: thousands

**Largest system and best logical-qubit record, by platform (log scale)**




**≈1,600×**

more physical qubits than the largest 2026 array

**≈60×**

more logical qubits, and far better ones

BELOW THRESHOLD

- Google Λ **2.14** · Dec 2024
- Harvard–MIT–QuEra **2.14×** · Nov 2025
- USTC Λ **1.40** · Dec 2025, reported

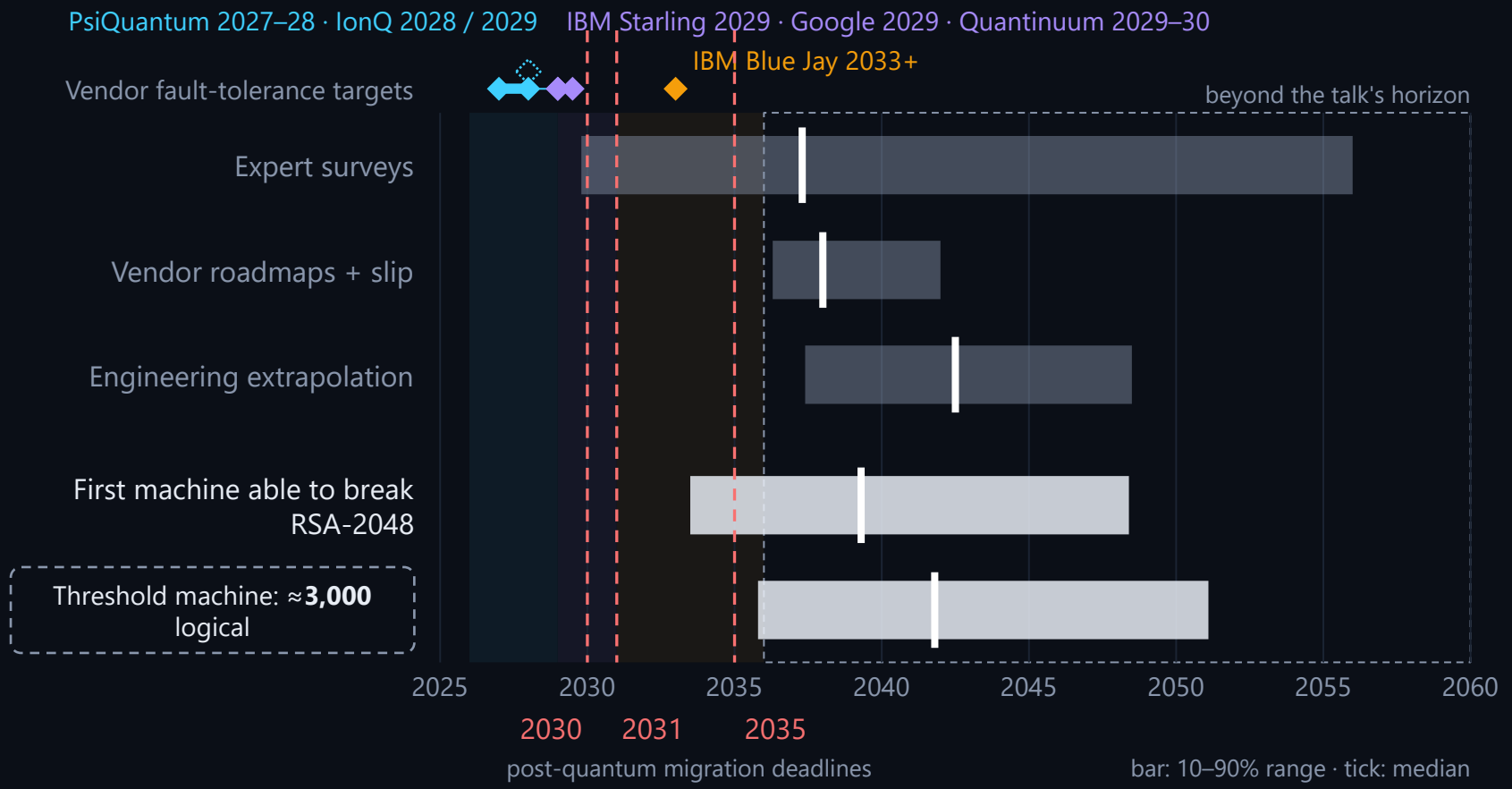
* error detection only

Platform records: IBM (Dec 2023) · Quantinuum (Nov 2025) · Caltech (Sep 2025) · Harvard/MIT/QuEra, Nature (Nov 2025) · Infleqtion (Sep 2026, vendor claim) · Xanadu, Nature (Jan 2025) · HRL, Nature (Jul 2026) · Google Willow, Nature (Dec 2024) · threshold machine: Qodeh model, Oct 2026



WHEN

# Roadmaps say 2029; the evidence says about 2040




**2029**

vendor roadmaps

**≈2040**

median · Qodeh model, Oct 2026

**Vendor targets, by year stated**



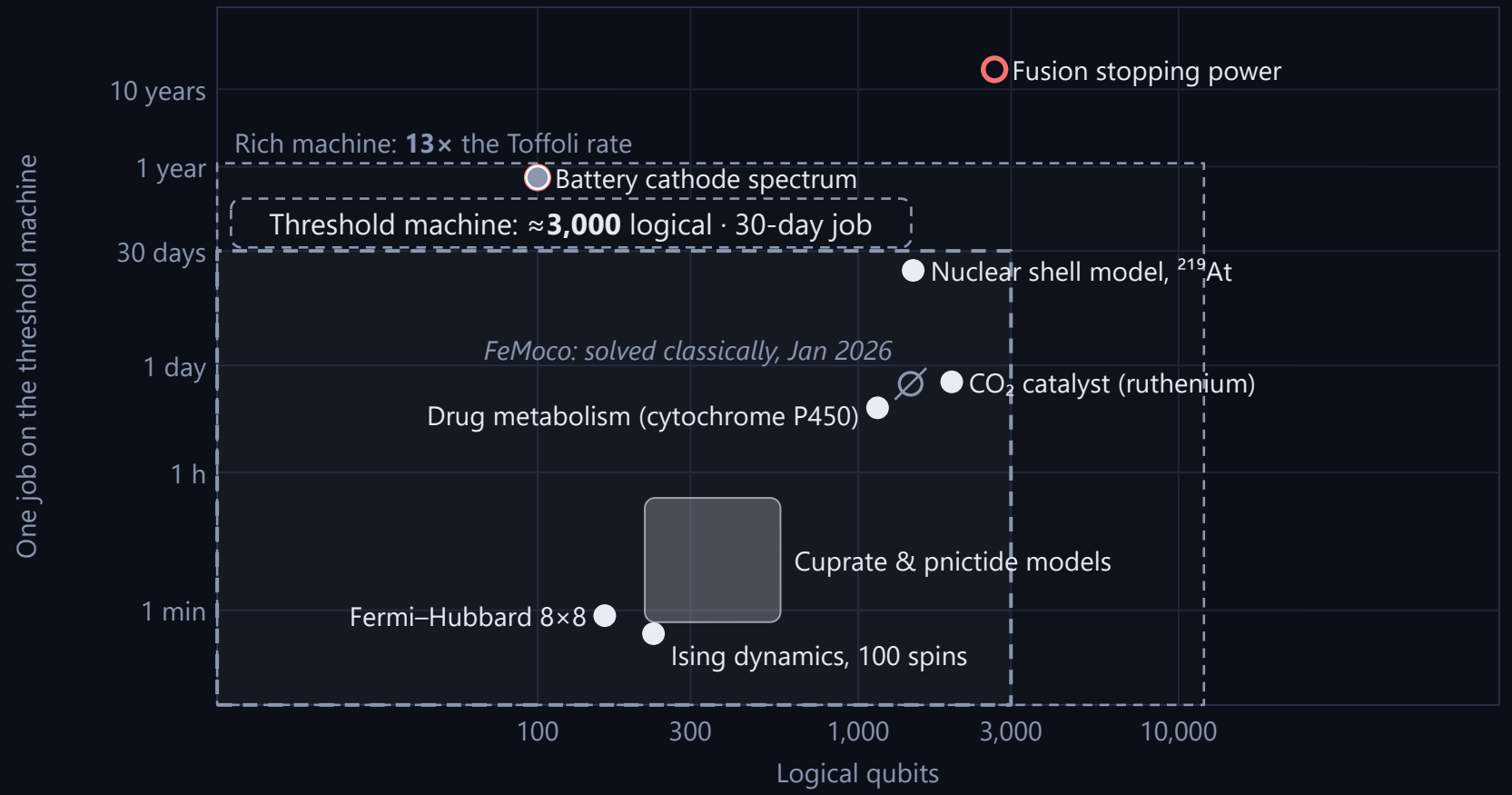

*always about 5 years away*




Qodeh model, Oct 2026 (QT-Map SP10, draft v2) · GRI Quantum Threat Timeline 2025 · IBM · IonQ, Quantinuum, Google, PsiQuantum roadmaps · US EO 14412 · NSM-10, UK NCSC, EU NIS CG

WHAT FOR

# A science instrument: molecules, materials, nuclei




**6 of 8**

core applications fit

Qodeh model, Oct 2026

## The Toffoli rate, not the qubit count, is the axis that pays.

Qodeh model, Oct 2026 (QT-Map SP10, draft v2), on published resource estimates: arXiv 2502.15882 · 2506.15784 · 2012.09238 · 2411.02160 · 2211.07629 · 2607.21563 · 2308.12352 · FeMoco classical: Zhai et al., arXiv 2601.04621

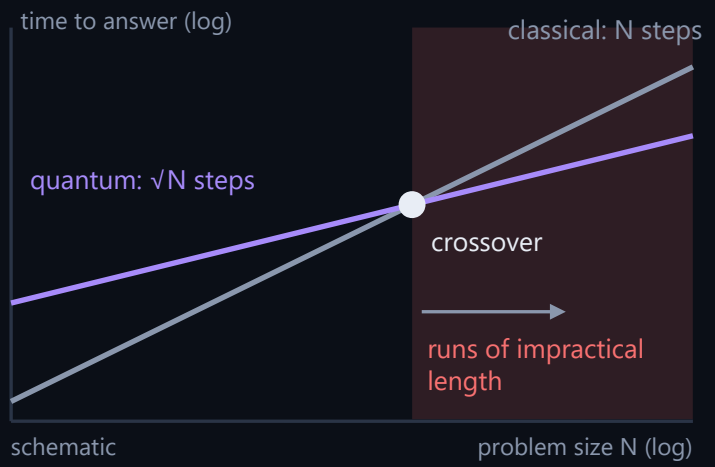
WHAT NOT FOR

# Quadratic speedups don't survive error correction

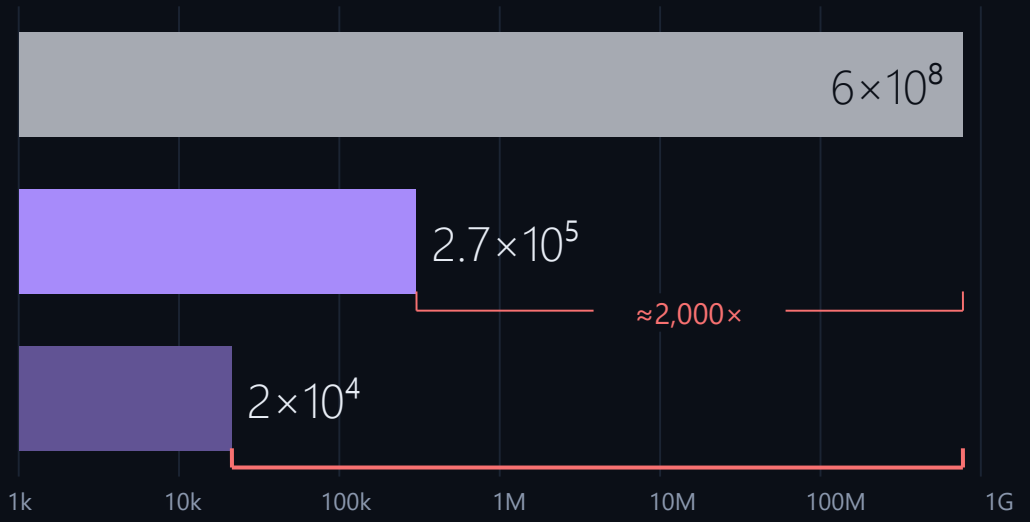
**1 ns**
one classical operation

**≥ 1,000×**
slower per step

**1 µs – 1 ms**
one error-corrected logical operation




**A derivative price in 1 s: Toffoli gates per second**




"45 MHz" = a T-depth rate, not throughput

**30,000× too slow**

**Out:** ~~Grover search~~ · ~~amplitude-amplified optimisation~~ · ~~Monte-Carlo pricing~~

Babbush et al., PRX Quantum 2021 · Hoefler, Häner & Troyer, CACM 2023 · Stamatopoulos & Zeng, Quantum 2024 · Qodeh model, Oct 2026

2026 — 2031 — 2036 → 2040s

QUANTUM AND AI

# So far, AI helps quantum more than the reverse

## AI → QUANTUM

- **Neural decoders** — 30% fewer errors
- **Calibration** — 3.7× lower error rates
- **Circuit compilation** — T gates fewer
- **Machine-found proofs** — 372 results

## QUANTUM → AI

- **Speed-ups dequantised** — 2018
- **Data loading (QRAM)**
- **Learning from quantum data** — 40 qubits
- **Quantum-made training data**

Not a fair test — yet

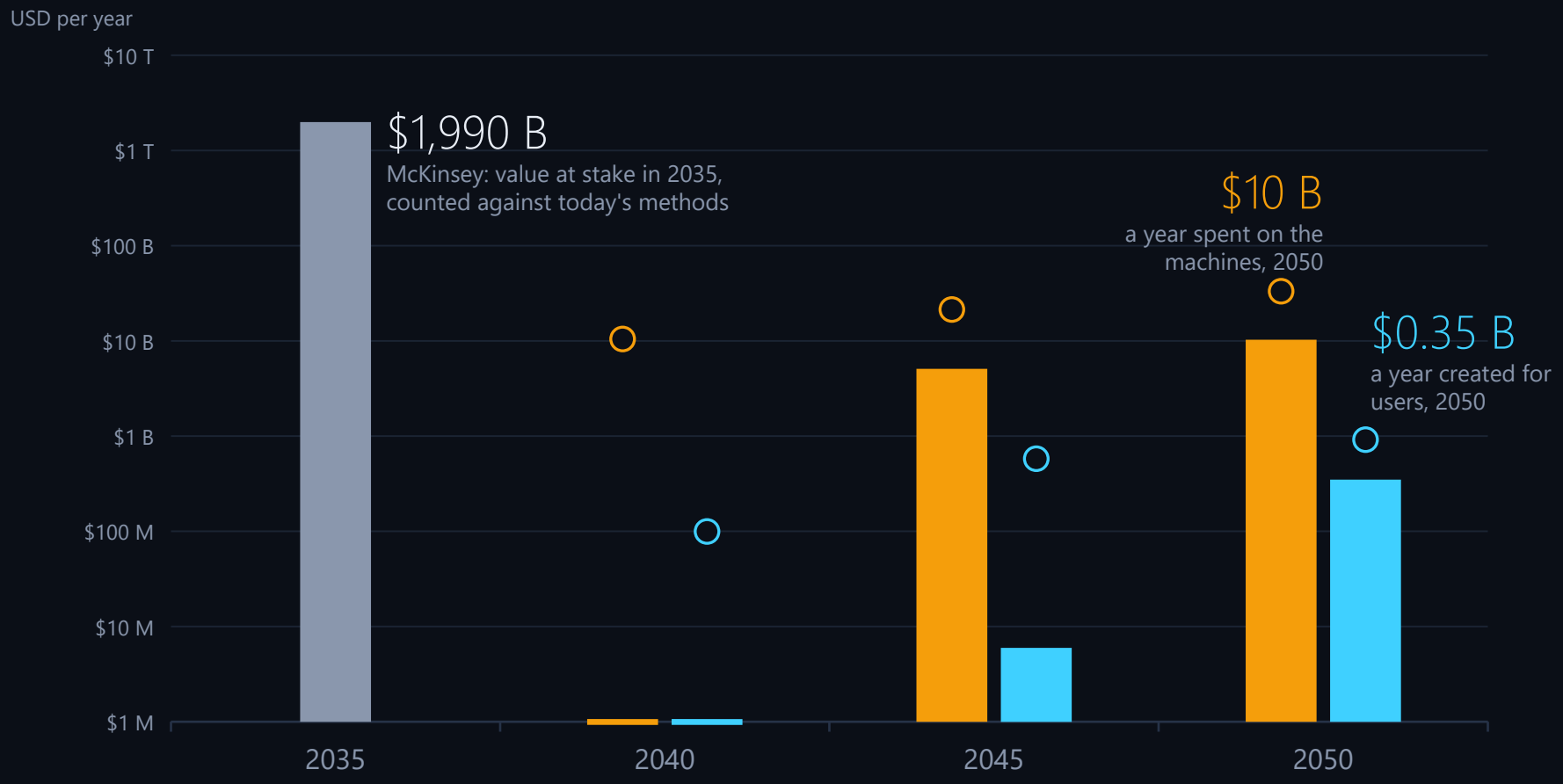
• refereed • preprint or unrefereed • blocked

Google DeepMind & Google Quantum AI, Nature 2024 · Klimov et al., Nat. Commun. 2024 · Ruiz et al., Nat. Mach. Intell. 2025 · Tang 2018 · Jaques & Rattew, Quantum 2025 · Huang et al., Science 2022 · Preskill, Q2B 2025



THE MONEY

# From $2 trillion to a third of a billion




**×30**

spent on the machines per dollar created, 2050

**$4.2 B**

all value created 2026–2070, present value (median)

Hundreds of millions a year, not trillions: **real money for a scientific instrument, not an economy.**

value at stake vs today's methods (McKinsey) · spent on the machines: capital, operation, software (vendors' revenue) · value created for users: time gained × value, vs same-era classical + AI

○ = P90: nine of ten simulated futures come in below it · — = median 0 (no machine yet) · Qodeh model, Oct 2026

McKinsey Quantum Technology Monitor 2026 · Qodeh model, Oct 2026 (QT-Map SP10, draft v2)



WHO PAYS

# Forty cents of every quantum dollar buy control electronics

Total market ≈ $10 B a year by 2050, median

**Software and services (add-on)**

| Application services | Software | Cloud access |
|---|---|---|
| 40% | 33% | 26% |

**Machines (fleet spend)**

| Control electronics | Cryogenics | Facility and power | Quantum processor | 8% |
|---|---|---|---|---|
| 40% | 20% | 20% | 12% | Decoding |

Who pays for the machines: state ≈ **99%** civilian ≈ **1%**

Qodeh model, Oct 2026 · splits are judgments

40%

control electronics

48% with decoding



2026 2031 2036

→ 2040s · beyond the talk's horizon

Qodeh model, Oct 2026 (QT-Map SP10, draft v2) · Gartner IT spending forecast, Feb 2026

THE LOOP

# The real-time loop is an FPGA problem

measure → decode → feed forward, every ≈ **1 µs**

| Quantum processor | Control electronics | Real-time decoder | Classical host (GPU, CPU) |
|---|---|---|---|
| | 40% | 8% | |
| round 1.1–1.7 µs | readout → decoder 1.4 µs | CPU software 63 µs | NVQLink round trip 3.84 µs |
| | | FPGA 0.44–0.79 µs/round | GPU 67 µs: ion-trap rounds only |
| | | IBM FPGA 40 ns/cycle, synthetic data | |

feed forward

**9.6 µs** for 9 rounds, end to end

## ≥ 1 MHz

decode rate, every round

Rigetti × Riverlane, arXiv:2410.05202 · Google, arXiv:2408.13687 · IBM, arXiv:2510.21600 · NVIDIA, Nov 2025 · Real-time QEC stack, arXiv:2605.30765



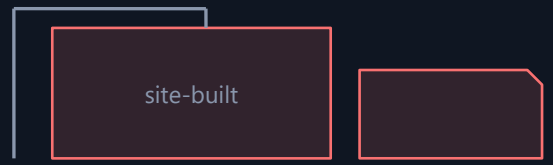
ARCHITECTURE

# Modular machines arrive six years sooner

**Granular: identical modules, factory-built**
fleet doubles every **1.5–2.5** yr · cost −**15–25%** per doubling

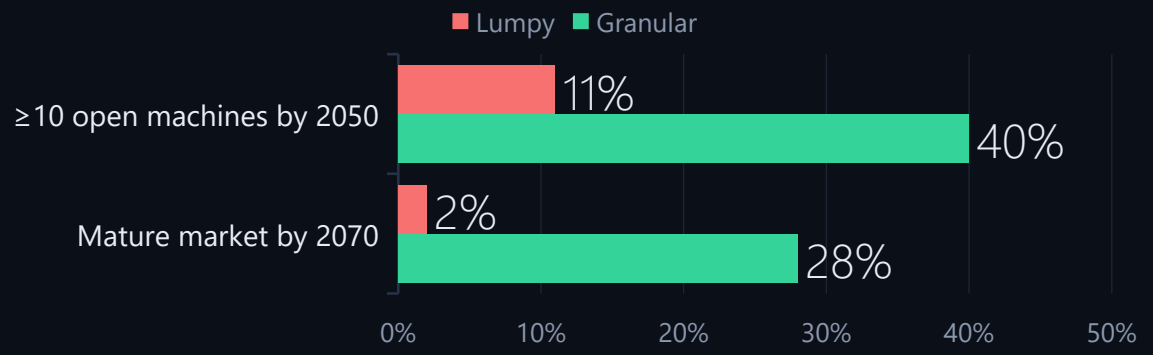



**Lumpy: bespoke, site-built**
every **3–5** yr · −**0–10%**










# 6 years

earlier ten-machine open fleet (median)

*Qodeh model, Oct 2026*




Qodeh FTQC value model (QT-Map SP10, draft v2, Oct 2026) · Wilson et al., Science 2020 · Grubler, Energy Policy 2010 · Li-ion costs (Joule 2022)

ISRAEL

# Israel: everything around the qubit — and now the qubit

**QUBITS**
five platforms

- Quantum Art — trapped ions
- Quantum Source — photonic qubits
- QuamCore — superconducting qubits
- new since 2025 — Q-Factor — neutral atoms
- Quantum Transistors — diamond colour centres

**CONTROL & DECODING**
pulses, timing, real-time decoding

- Quantum Machines
- new since 2025 — QuantumXLabs

**SOFTWARE & ERROR MITIGATION**
compilers, noise cancellation

- Classiq
- Qedma
- QuantyMize — Quantum advance
- Qarakal

**PHOTONICS & SECURE LINKS**
crystals, quantum key distribution

- Raicol
- HEQA Security

**QUANTUM-INSPIRED**
laser computing, no qubits

- LightSolver

$240M
Qubit builders

$170M
Control

$136M
Software

*verified rounds, Sep 2024–Apr 2026*

2026 2031 2036

Company releases and TechCrunch, Globes, QCR, TQI (Feb 2025–Apr 2026) · Qodeh roster, Oct 2026

OPEN QUESTIONS

# The open questions of the theory

1

## Can it crack NP-complete problems?

Search: at most √N faster (BBBV 1997)

open since 1997

2

## Does nature allow more than quantum?

Nonlinearity or time loops would (Abrams–Lloyd; Aaronson–Watrous)

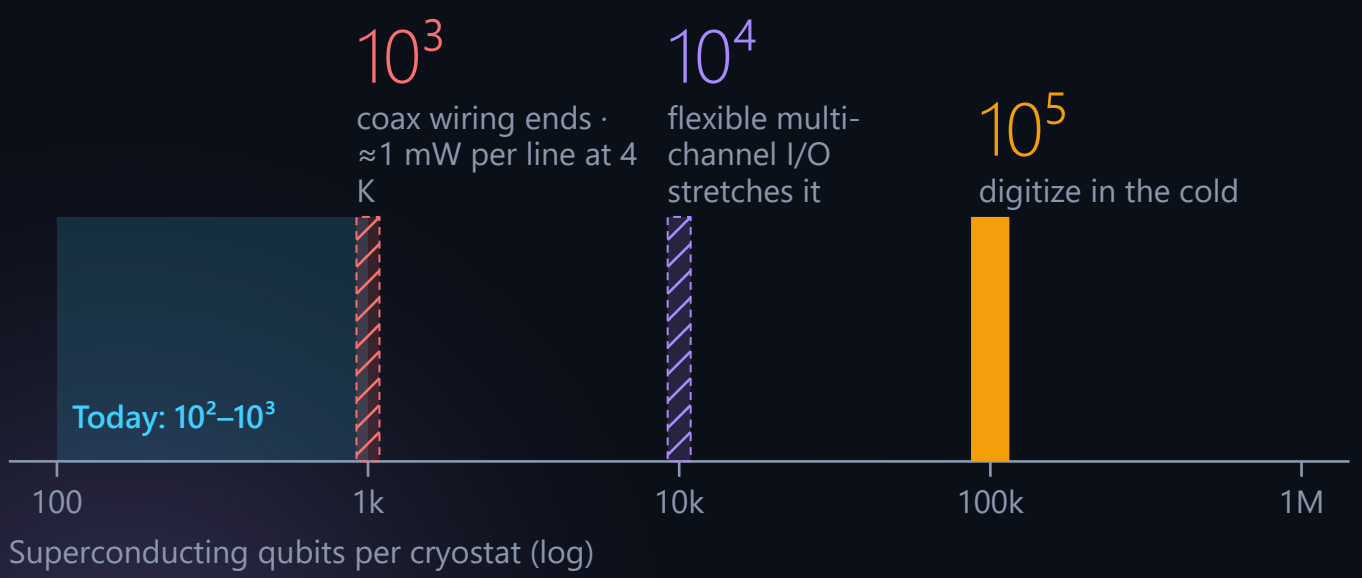
open since 1998

Bennett–Bernstein–Brassard–Vazirani 1997 · Abrams–Lloyd 1998 · Aaronson–Watrous 2009

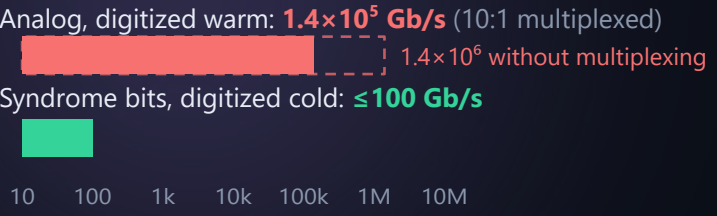


SUPERCONDUCTOR ELECTRONICS

# Past 100,000 qubits, digital logic moves into the cold

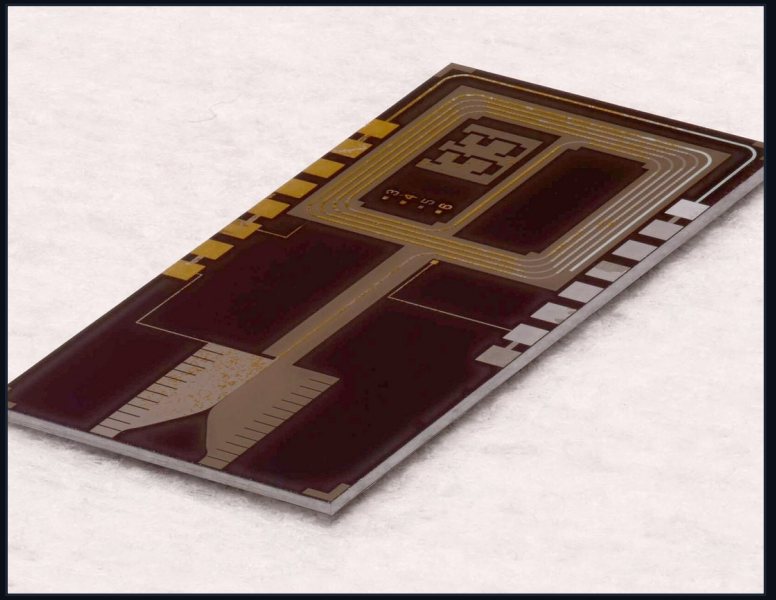



**Data leaving the fridge, $10^5$ qubits, 1 µs round**




Cryo-CMOS 23 mW/qubit (4 K)

SFQ control nW/qubit (10 mK)



*"No stack demands SFQ today; the demand emerges with scale, is specific to the superconducting modality, and — on demonstrated evidence — lands on SFQ first when it does."*

Qodeh SFQ–HPC bridge analysis, Aug 2026

2026 2031 2036

Krinner et al., EPJ Quantum Technol. 2019 · Underwood et al., PRX Quantum 2024 · SEEQC, Nature Electronics 2026 · Bluefors–Delft Circuits 2025 · Qodeh analysis

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